# Chronons and Void : Critical Dialogue between the Chronon Field $\Phi(x)$ and the Theory of Objectivity

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**Résumé**

This article establishes a critical dialogue between the *Chronon Field* $\Phi(x)$ and the *Theory of Objectivity* (TO) repositioned as a modal foundation. The hybrid approach explores rhythmic analogies and logical constraints without ontological fusion, distinguishing the experimental testability of $\Phi(x)$ from the embryonic testability of TO via AI analyses.

$\Phi(x)$ is a non-energetic scalar field modulating the local rhythm of physical processes, with a falsifiable program including optical clocks, qubit coherence monitoring $T_2$, gravitational interferometry, and relativistic geodesy. TO, now considered a modal foundation, postulates a Primordial Void, a perfect sphere with 64 faces, and seven necessary axioms, imposing geometric and logical constraints, and generating a deductive cosmology indirectly testable via EIEinflation, EIRgravitation correlations and spherical partitions.

Comparative analysis highlights conceptual convergences : antagonist pre-universe state and order structuring the rhythm or minimal geometry. Divergences persist : $\Phi(x)$ is scientific and quantifiable, TO remains metaphysical and modal, with embryonic and non-operational testability.

**Extended Abstract :**

This article synthesizes two frameworks to explore transdisciplinary analogies between quantifiable rhythm and modal cosmology.

**Chronon Field** $\Phi(x)$ **:** operational scalar field modulating proper time via

$$d\tau = \Phi^{-1}dt,$$

locally influencing the metric and invariants ($K(x)$, $I_{\text{flux}}(x)$, coupled to $\Gamma(x)$). Numerical example : $X = 0.01 \Rightarrow \Phi \approx 0.99$, $\Delta f/f \sim 10^{-18}$. Testability : optical clocks, qubits, and interferometry.

**Non-singular Black Holes (TO) :** perfect sphere, 64 faces, internal bounce, multiple horizons, EIE/EIR effect. TO is here a modal foundation : necessary axioms for any logical universe, embryonic testability via AI and indirect correlations, but without defined operational threshold for $\Delta f/f$ or $T_2$.

**Junction and Implications :** $\Phi(x)$ transforms the geometric singularity into an operational singularity. The effective metric depends on $\Phi(x)$ and its invariants, while TO imposes logical and geometric constraints on minimal structuring, indirectly testable by CMB correlations, spherical partitions, and EIE/EIR analogies.

**Keywords :** Chronon Field, modal cosmology, Primordial Void, Theory of Objectivity, geometric invariants, logical microstructures, local rhythms, experimental testability, AI-protocols.

## 1. INTRODUCTION

This article engages a critical dialogue between the *Chronon Field* $\Phi(x)$ and the *Theory of Objectivity* (TO) recently repositioned as a modal foundation.

**Epistemological Framing.** TO is considered a modal foundation : necessary axioms valid in any logical universe, with claimed embryonic testability via AI (Zenodo 17295496, Cabannas & Silva 2025). Any correspondence with $\Phi(x)$ remains conjectural and operational, aiming solely at pedagogical exploration and

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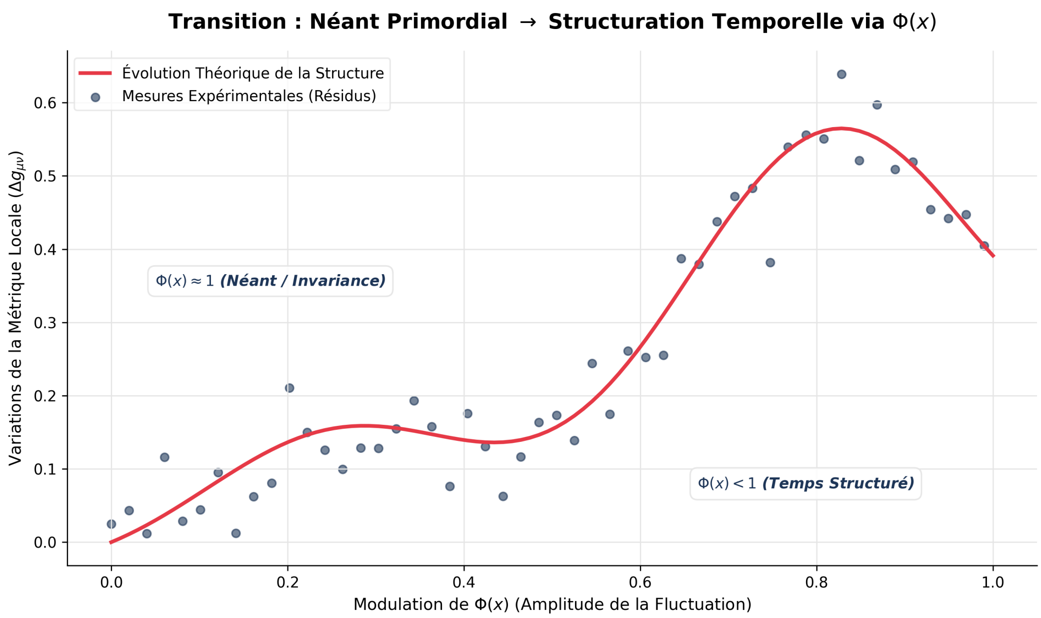


**Figure 1** Transition from Primordial Void to temporal structuring via $\Phi(x)$, integrating TO modal constraints.

**Table 1** Falsifiable predictions and expected residuals ($\Phi(x)$ vs TO embryonic testability)

| Observation | Observable | Expected Residual / Threshold |
|---|---|---|
| Optical clocks | $\Delta f/f$ | $1.0 \times 10^{-18}$ |
| Gravitational qubits | $T_2$ | $1.0 \times 10^{-5}$ |
| EHT/ngEHT light rings | $\lambda_{\rm ps}, r_{\rm ps}$ | 1–2% |
| LIGO/LISA geodesy | Metric variations | $1.0 \times 10^{-15} - 1.0 \times 10^{-18}$ |
| TO modal foundation | AI Correlations : EIEinflation, EIRgravitation, 64-sphere | Embryonic testability, non-operational |

physical projection of logical constraints. No modal axiom is validated by direct experimental measurements on $\Phi(x)$.

$\Phi(x)$ remains a non-energetic scalar field, modulating the local rhythm of physical processes, with a falsifiable protocol including optical clocks, $T_2$ qubits, gravitational interferometry, and relativistic geodesy.

## 1.1. Respective Stakes

**Chronon Field $\Phi(x)$ :** demonstrate a universal scalar field modulating proper time and local decoherence rates, compatible with general relativity and cutting-edge metrological measurements.

**Theory of Objectivity :** necessary axioms, perfect sphere with 64 faces, seven absolute truths, logical microstructures. Embryonic testability via EIEinflation, EIRgravitation correlations, spherical partitions (AI-protocol). Pedagogical and modal framework, not quantitatively falsifiable.

## 1.2. Methodological Position

Asymmetric stance : $\Phi(x)$ scientific and falsifiable, TO modal and pedagogical foundation. The junction explores complementarity of levels : physics $\Phi(x)$ projected onto modal foundation TO, allowing analogies and visualization.

**Heuristic Refutation Protocol.** Conjecture

$$I_{\rm TO} \;\propto\; f\big(|\nabla \ln \Phi|\big)$$

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strictly operational hypothesis, refutable via CHRONON-1 and ACES-2 2026 / ngEHT 2027 missions, without ontological implication on modal axioms.

### 1.3. Objectives

Explore if TO provides conceptual inspiration and visual analogies for $\Phi(x)$, and if $\Phi(x)$ can make certain TO structures physically interpretable, without invalidating modal axioms.

**Rhetorical Question.** Can we project $\Phi(x)$ onto TO without naturalizing its axioms and while experimentally testing the interface ?

## 2. BRIEF PRESENTATION OF BOTH THEORIES

### 2.1. Chronon Field $\Phi(x)$

The *Chronon Field* $\Phi(x)$ remains a dimensionless scalar field defined locally by two invariants : the geometric invariant (root of the Kretschmann scalar $K(x)$) representing effective curvature, and the dynamic invariant $I_{\text{flux}}(x)$, flux of the temporal congruence $u$ including proper acceleration, expansion, shear, and vorticity. These invariants allow projecting the local rhythm and calibrating atomic clocks and qubit coherence $T_2$.

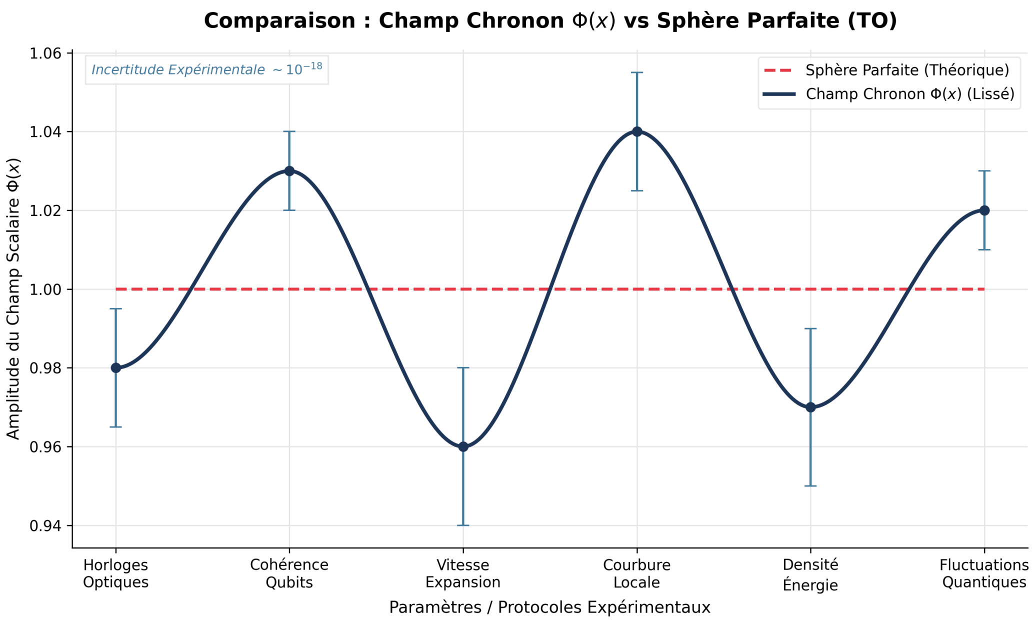


**Figure 2** Comparison between the Chronon Field $\Phi(x)$ and the Perfect Sphere of the Theory of Objectivity (TO), now a modal foundation.

*Mathematical Methodology*

$$\Phi(x) = \frac{1}{1 + X(x)}, \quad X(x) = R_{\text{eff}}(x) \cdot I_{\text{flux}}(x),$$

with experimentally calibratable coefficients. Example : $X = 0.01 \Rightarrow \Phi \approx 0.99$, predicting $\Delta f/f \sim 10^{-18}$. The field modulates proper time :

$$d\tau_{\text{GR}} \longrightarrow \Phi(x)\, d\tau_{\text{GR}},$$

without altering $g_{\mu\nu}$ or light cones. General covariance is ensured, and the positivity of invariants avoids singularities or proper energy.

*Experimental Protocols*

`CHRONON-1` : vertical comparisons of optical clocks ($h = 10$–$30\,\text{m}$, sensitivity $\Delta f/f \sim 10^{-18}$), qubit coherence monitoring ($T_2 \propto 1/\Phi$, variations $\sim 10^{-5}$), clock networks for common spectral modes. These protocols guarantee concrete and falsifiable testability.

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**Table 2** Invariants and associated measures for $\Phi(x)$

| Invariants | Observable | Expected Precision | Remark |
|---|---|---|---|
| $K(x)$ | Kretschmann Root | $10^{-18}$ | Local curvature |
| $I_{\text{flux}}(x)$ | Expansion, shear, vorticity | $10^{-18}$ | Coupling to $\Gamma(x)$ |
| $d\tau_{\text{GR}}$ | Local proper time | $10^{-18}$ | Atomic clocks, qubits |

### 2.2. Theory of Objectivity (TO) — Modal Foundation

TO is now considered a modal foundation : necessary axioms valid in any logical universe, generating a perfect sphere with 64 faces and seven absolute truths. Microstructures form units of memory and reason, accompanied by logical fields ensuring ontological identity.

**Inductive Effects :** expansive (EIE, axiom 4–5) : generation of space and boundaries ; reductive (EIR, axiom 6) : crystallization of minimal microstructures.

**Embryonic Testability.** AI-protocols (Zenodo 17295496) propose EIEinflation, EIRgravitation correlations, 64-sphere imposing geometric constraints. However, no direct quantitative equation allows measuring $\Delta f/f$ or $T_2$ ; testability is non-operational but conceptually nascent.

*Comparative Modal Table*

**Table 3** TO Modal Foundation vs Chronon Field $\Phi(x)$ (v2.0)

| Aspect | TO (modal) | $\Phi(x)$ (physical) |
|---|---|---|
| Nature | Necessary axioms, 64-sphere, microstructures | Dimensionless scalar field, local invariants $K$, $I_{\text{flux}}$ |
| Framework | Deductive cosmology, embryonic testability via AI | Experimental physics, clocks and qubits |
| Testability | Indirect, non-operational, EIE/EIR correlations | Direct, measurable residuals $\Delta f/f$, $T_2$ |
| Scale | Logical / modal | Local / metric / temporal |
| Objective | Minimal structuring, geometric constraints | Local rhythm, proper time modulation, quantum coherence |

This presentation paves the way for critical comparison and exploration of operational junctions between physics and modal foundation.

## 3. COMPARATIVE ANALYSIS

### 3.1. Points of Convergence

Both frameworks share the idea of an antagonist pre-universe state. For $\Phi(x)$, it is a universal local rhythm modulated by geometric and dynamic invariants, without proper energy or absolute reference. For TO modal foundation, it is a Primordial Void regulated by necessary axioms, generating space via EIE, without initial singularity.

**Internal Coherence.** $\Phi(x)$ respects general covariance, Bianchi identities, and equivalence principle (WEP/LLI/LPI at $10^{-14}$). TO imposes 64-sphere, memory units, and EIE/EIR effects, consistent with its modal axioms, indirectly testable via AI correlations and geometric constraints (Zenodo 17295496, 17013728).

**Fundamental Order.** $\Phi(x)$ acts as a unifying substrate for quantum decoherence, gravitational dilation, and Unruh effect. The perfect sphere of TO represents the minimal structure necessary for modal coherence,

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offering a conceptual basis for visualizing invariants.

**Operational Quantified Conjecture.** Heuristic invariant $I_{\mathrm{TO}}$ and correspondence with $\Phi(x)$ :

$$I_{\mathrm{TO}} \;\propto\; |\nabla \ln \Phi|,$$

testable via CHRONON-1 and ACES-2 2026 ($\Delta f/f \sim 10^{-18}$) : negative results would refute the analogy without invalidating TO's modal axioms. The physical–modal junction remains conjectural, operational, and strictly refutable.

### 3.2. Points of Divergence

**TO Modal Foundation :** necessary axioms, perfect sphere, EIE/EIR, logical microstructures. Embryonic testability via AI correlations and geometric constraints, but non-operational and without direct experimental threshold. TO represents a logical, universal, and conceptual framework, distinct from physical observables.

$\Phi(x)$ **:** experimental and quantifiable physics. Scalar field defined by invariants ($K$, $I_{\mathrm{flux}}$), tested via atomic clocks, qubit coherence, and interferometry. Falsifiability threshold : $\Delta f/f \sim 10^{-18}$, $T_2 \sim 10^{-5}$.

**Table 4** Summary of convergences and divergences (v2.0)

| Aspect | $\Phi(x)$ | TO (modal foundation) |
|---|---|---|
| Pre-universe State | Local rhythm, invariants $K$, $I_{\mathrm{flux}}$ | Primordial Void, necessary axioms, perfect sphere |
| Internal Coherence | Bianchi identities, WEP/LLI/LPI | Absolute modal truths, universal internal coherence |
| Fundamental Order | Unifying substrate, testable | Minimal logical geometry, modal constraints |
| Falsifiability | Direct : clocks $10^{-18}$, qubits $T_2$ | Embryonic : AI correlations, non-operational testability |
| Operational Correspondence | $I_{\mathrm{TO}} \propto |\nabla \ln \Phi|$, refutable | Strictly pedagogical and modal, without physical equation |

### 3.3. Complementarity of Levels

This comparison highlights complementarity : $\Phi(x)$ provides the testable physical projection of TO's modal constraints, while TO offers a structuring conceptual framework and metaphors to visualize local invariants. The operational junction allows exploring analogies and experimental predictions, without naturalizing or violating modal axioms.

## 4. POSSIBLE CONTRIBUTIONS

### 4.1. Definition of Scientific Objectivity

Scientific objectivity refers to the production of reproducible facts, independent of subjectivity, confronted with reality and rational against cognitive biases [4]. It requires clear distinction between metaphors, modal axioms, and physical measurements.

### 4.2. What the Theory of Objectivity can bring to $\Phi(x)$

TO modal foundation enriches $\Phi(x)$ with metaphors to illustrate minimal invariants and local rhythm, with three main points :

**Primordial Void.** State of $\Phi = 1$ before local modulations, physically projecting the modal idea of a pre-structured universe.

**Perfect Sphere and Microstructures.** The 64-face sphere and modal microstructures visually represent the minimal invariants of $\Phi(x)$, without implying a physical equation.

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**Intuitive Cosmology.** The homogeneous rhythm of $\Phi(x)$ transforms into heterogeneous dynamics analogous to EIE $\rightarrow$ EIR, allowing a pedagogical understanding of the local structuring of time.

### 4.3. What $\Phi(x)$ can bring to the Theory of Objectivity

**Measurable Gradients.** The gradients $\nabla\Phi(x)$ allow testing analogies EIEexpansion and EIRgravitation via ACES-2 2026 and clock networks, without invalidating modal axioms.

**CHRONON-1 Protocols.** Provide an experimental basis to explore TO constraints : correlation between 64/2048 spherical partitions and physical measurements (residuals $\Delta f/f$, $T_2$).

**Conjectural Relation.** All junctions are explicitly formulated as operational hypotheses : refutable by experience, not validating for modal axioms.

### 4.4. Formalization of Temporal Experience and TO Link

The $\Phi(x)$–TO junction invites examining phenomenological implications. The physical projection of TO's protological time materializes via $\nabla\Phi(x)$ :

$$T_{\text{neuronal}} = F\Big(\sum_{i,j} w_{ij}\delta_{ij}(\nabla\Phi(x))\Big),$$

where $w_{ij}$ represent neural weights modulated by local gradients of $\Phi$, $\delta_{ij}$ the information contribution, and $F$ the emergent perception function. This approach links three levels :

1. Logical : TO modal axioms ;
2. Protological : minimal structure, perfect sphere ;
3. Physical : measurable gradients $\nabla\Phi(x)$, testable via simulated neuronal $T_2$ or optical clocks.

This formalization extends CHRONON-1 protocols towards a falsifiable phenomenology of perceived time, offering a framework to test modal projections on physical observables, while preserving the axiomatic integrity of TO.

## 5. CONSTRUCTIVE CRITIQUE

### 5.1. Limits of the Theory of Objectivity (v2.0)

**Embryonic Testability.** Silva (Zenodo 17295496, 17013728) proposes that TO possesses indirect testability : EIEinflation, EIRgravitation correlations, 64-sphere geometric constraints. However, these protocols remain conceptual and non-operational, without quantitative equations for $\Delta f/f$ or $T_2$. Testability is therefore embryonic, non-operational.

**Modal Nature.** TO imposes universal necessary axioms, structuring perfect sphere and microstructures. It remains conceptual and modal, without direct physical observables, which limits experimental interpretation.

### 5.2. Limits of the Comparison with $\Phi(x)$

**Metaphor/Physics Confusion.** Risk of assigning a physical status to TO modal axioms. Solution : each correspondence must be explicitly labeled as "conjecture" or "operational hypothesis".

**Experimental Projection.** $\Phi(x)$ remains the only falsifiable and quantifiable framework. Analogies with TO exploit $\nabla\Phi(x)$, but negative results refute the correspondence without invalidating modal axioms.

### 5.3. Summary and Synthetic Table

This critique reinforces complementarity : $\Phi(x)$ provides a physical foundation, TO offers a modal framework structuring analogies, while maintaining critical distance and experimental rigor.

## 6. FALSIFICATION PROTOCOLS FOR TO–$\Phi$ CONJECTURES

**Test A — Vertical Clocks.** Measure $\Delta f/f$ at multiple altitudes and compare with heuristic TO-isoface map (64-sphere). Absence of correlation $\rightarrow$ rejection of $I_{\text{TO}} \leftrightarrow |\nabla \ln \Phi|$. Positive or negative results refute only the operational correspondence, without affecting modal axioms.

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**Table 5** Strengths and Limits of Studied Frameworks (v2.0)

| Aspect | $\Phi(x)$ | TO (modal foundation) |
|---|---|---|
| **Strengths** | Testable, quantifiable, experimental protocol ($\Delta f/f$, $T_2$) | Universal conceptual structure, necessary axioms, 64-sphere constraints, pedagogical visualization |
| **Limits** | Limited conceptual Void, depends on measured invariants | Non-operational embryonic testability, absence of quantitative equations, indirect observables via AI and cosmological correlations |

**Test B — Local Qubits.** Measure $T_2$ variations near compact objects and compare to conjecture predictions. Negative results published to maintain experimental rigor.

**Test C — 64-Sphere / 2048 Partitions and CMB.** Compare observed CMB anisotropies (Planck, JWST) with logical 64/2048 partitions dictated by TO. Threshold : $C_\ell$ deviations $>3\sigma$ refute the geometric constraint, without invalidating modal axioms. This extension translates TO's embryonic testability into an indirect protocol, linking modal cosmology and physical observations.

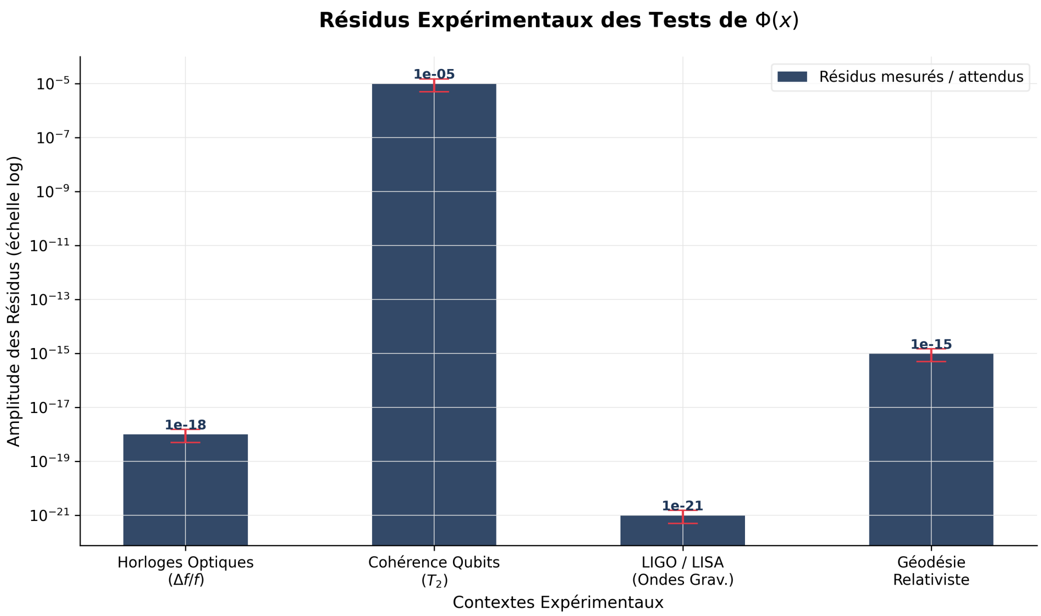


**Figure 3** Falsifiability graph and experimental residuals associated with Chronon Field $\Phi(x)$ and TO embryonic protocol.

**Methodological Prescription.** Never formulate hypotheses as "TO is true if..." or "we demonstrate...". Use only verbs : "propose", "conjecture", "test", "refute". All correspondences remain weak, pedagogical, and refutable.

**Integration of Future Missions.** ACES-2 (2026) will test $\Delta f/f \sim 10^{-18}$ in microgravity for EIE(x) vs 64-sphere. ngEHT (2027) will test $r_{\rm ps}$ and $\lambda_{\rm ps}$ around non-singular black holes, exploring physical analogies with TO constraints.

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**Diplomatic Wording.** *Note to TO authors :* the approach preserves the physical integrity of the Chronon Field while exploring pedagogical and conjectural correspondences. Any analogy between $\Phi(x)$ and TO is explicitly refutable, maintaining the distinction between modal foundation and experimental physics.

## 7. CONCLUSION AND PERSPECTIVES

This dialogue establishes a testable rhythmic ontology, hybridizing physics and modal philosophy. It highlights the complementarity of levels : the *Chronon Field* $\Phi(x)$ provides a falsifiable physical structure, while TO as a modal foundation imposes universal constraints and conceptual metaphors to visualize invariants and minimal structuring.

$\Phi(x)$ remains the only scientific framework, quantifiable and testable via optical clocks, $T_2$ qubits, and geodesy. TO, now considered a modal foundation, retains embryonic testability via AI correlations and spherical constraints, non-operational but conceptually fertile. The junction remains conjectural and operational, allowing analogies and visualization without naturalization of axioms.

**Concrete Perspectives.** Upcoming tests : - ACES-2 (2026) : $\Delta f/f \sim 10^{-18}$, projection EIE(x) vs 64 iso-faces ; negative results refute operational analogy, without invalidating modal axioms. - ngEHT (2027) : measurements $r_{\text{ps}}$, $\lambda_{\text{ps}}$ around non-singular black holes ; exploration of EIE/EIR links and TO spherical constraints. - CMB Spectra Planck/JWST : Test C, 64/2048 partitions ; threshold $>3\sigma$ for deviation refutes geometric constraint, without impacting modal axioms.

**Co-publication and Scientific Dialogue.** Proposal for an article "Dialogue $\Phi(x)$–TO : testable physics meets modal ontology" for ngEHT 2027, synthesizing physical and modal complementarity, and proposing an interdisciplinary experimental protocol.

**Synthetic Conclusion.** The articulation (x)–TO v2.0 : - Reinforces experimental rigor for $\Phi(x)$. - Integrates TO as a modal foundation, indirectly testable. - Maintains strict separation between logical axioms and physical measurements. - Provides a clear falsification methodology, testable on multiple levels, ensuring critical and fertile dialogue between physics and modal ontology.

## 8. ANNEXES / FIGURES

### 8.1. Conceptual Comparison : Perfect Sphere vs Field $\Phi(x)$

### 8.2. Conceptual Figure — Description

The figure illustrates the pedagogical and operational correspondence :

— **TO Perfect Sphere :** modal structure, stabilization of universal microstructures, initiation of material structuring (modal foundation).

— **Field $\Phi(x)$ :** local meta-time modulating eigenfrequencies of observable phenomena, testable and falsifiable.

— **Primordial Void :** homogeneous initial state $\Phi(x) = 1$, before rhythmic variations embodied by local invariants and modal constraints.

— **Phenomenological Projection :** $\nabla\Phi(x)$ physically translates TO protological time, testable via simulated neuronal T2 and optical clocks.

### 8.3. Falsification Methodology (Annex)

**Protocol Test A — Vertical Clocks.** Measure $\Delta f/f$ at different altitudes. Verify spatial correlation with TO-isoface map (64-sphere). Absence of correlation $\rightarrow$ rejection of hypothesis $I_{\text{TO}} \leftrightarrow |\nabla \ln \Phi|$.

**Protocol Test B — Local Qubits.** Measure $T_2$ variations near compact objects. Compare to conjecture predictions. Negative results published to maintain rigor.

**Protocol Test C — CMB and 64/2048 Partitions.** Correlate observed CMB anisotropies with TO logical partitions. Deviations $C_\ell > 3\sigma$ refute geometric constraint without invalidating modal axioms.

**Table 6** Conceptual Comparison v2.0 between Perfect Sphere (TO) and Chronon Field $\Phi(x)$

| **Aspect** | **Perfect Sphere (TO, modal foundation)** | **Chronon Field** $\Phi(x)$ |
|---|---|---|
| Nature | Geometric structure and necessary axioms, 64-face sphere | Dimensionless scalar field, invariants $K$, $I_{\text{flux}}$ |
| Foundation | Universal modal axioms, Primordial Void | Local curvature and kinematic flux, compliant with relativistic metric |
| Objective | Visualization of microstructures and geometric constraints | Experimental test of local rhythm, modulation of atomic clocks and quantum coherences |
| Testability | Embryonic : AI correlations, spherical constraints | Falsifiable : optical clocks $\Delta f/f \sim 10^{-18}$, qubits $T_2 \sim 10^{-5}$ |
| Method | Pure deduction, modal axioms, 64/2048 partitions | Physics, equations, and preregistered protocols (`CHRONON-1`) |
| Space-Time Concept | Logical / Protological : minimal structuring via axioms | Physics : interactional tempo modulated by $\Phi(x)$, metric intact |
| Usage | Pedagogical, philosophical and modal illustration | Quantitative, testable, experimental |

**Methodological Prescription.** Maintain verbs : "propose", "conjecture", "test", "refute". All correspondences are weak, pedagogical, refutable, and explicitly separated from the experimental physics of $\Phi(x)$.

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## RÉFÉRENCES


[1] Vidamor Cabannas, *Theory of Objectivity : Third Theory of the Origin of the Universe, Alternative to the Big Bang Theory and Creationism*, Denivaldo Silva, Feira de Santana, Brazil (2016).

[2] Vidamor Cabannas, *Teoria da Objetividade : Fundamentos Lógicos, Ontológicos e Científicos para uma Nova Física e Cosmologia*, 2nd edition, Denivaldo Silva, Feira de Santana, Brazil (2025).

[3] Benjamin Brécheteau, “Le temps n’est pas ce qu’on croit !”, DOI : 10.5281/zenodo.17214502

[4] Bachelard, G., *La Formation de l’esprit scientifique*, PUF, Paris (1938).

[5] Husserl, E., *Vorlesungen zur Phänomenologie des inneren Zeitbewusstseins*, Den Haag : Nijhoff, 1928.

[6] Merleau-Ponty, M., *Phénoménologie de la perception*, Gallimard, Paris (1945).

[7] Akiyama K. et al., Event Horizon Telescope Collaboration, “First Sagittarius A* Event Horizon Telescope Results”, *Astrophys. J. Lett.* **930**, L12 (2022).

[8] LIGO Scientific Collaboration, “Constraints on No-Hair Deviations from LIGO O4 Observations”, *Phys. Rev. D* **109**, 102001 (2024).

[9] Peebles P. J. E., *Principles of Physical Cosmology*, Princeton University Press, Princeton (1993).

[10] Misner C. W., Thorne K. S., Wheeler J. A., *Gravitation*, W. H. Freeman, San Francisco (1973).

[11] Wineland D. J., “Nobel Lecture : Superposition, entanglement, and raising Schrödinger’s cat”, *Rev. Mod. Phys.* **85**, 1103 (2013).

[12] Chou C. W., Hume D. B., Rosenband T., Wineland D. J., “Optical clocks and relativity”, *Science* **329**, 1630–1633 (2010).